TANIA CLAYTON

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EDUCATION

Master of Economics (Applied Track) | National University of Singapore — Aug 2026 – Jul 2028
- Awarded merit-based rebate by the Department of Economics.

BA (Hons.) Economics | Madras School of Economics, Chennai — Aug 2023 – May 2026
- CGPA: 8.80/10; Teaching Assistant for Mathematical Methods II and Introduction to Python & Data Science.

CORE COMPETENCIES

Quantitative & Financial Analysis: Technical Analysis, Options Flow Analysis, Backtesting, Risk Management, Econometrics, Time Series Analysis

Data & Engineering: Python, Flask, SQL (MySQL), AWS, NLP & Sentiment Analysis, Web Scraping, Data Cleaning & Wrangling

Business Intelligence: Power BI (DAX, Power Query), Excel, Data Visualization, Dashboard Development

AI Fluency: Vibe Coding, Prompt Engineering, AI-Assisted Development, LLM-Powered Workflows, Claude Skills

WORK EXPERIENCE

Data Analyst Intern | Innova8s, Singapore — Sept 2026 – Present
- Designed UI and user flows for React Native poultry and cattle monitoring apps, applying behavioral science to help users read ground-level data and act on it.
- Curated multimodal visual datasets into annotation-ready pipelines (Roboflow) for AI/ML training and computer vision detection.
- Represented the company at a patent training program, engaging with IP experts on the company's behalf.

Quantitative Finance Intern | Janus Technology (Finance Wing), Chennai — May 2026 – Aug 2026
- Tasked with building a research-to-execution analytics pipeline; designed and deployed a full-stack trading platform (Python, Flask, MySQL, AWS) computing a proprietary RSI-based indicator across NSE, NASDAQ, and NYSE equities.
- Engineered a multi-factor signal suite blending technical indicators (VWAP, Bollinger Bands, momentum, mean reversion) with NLP-driven news and social sentiment analysis, producing backtested, confidence-weighted trade recommendations.
- Applied platform-generated signals to execute and manage intraday and swing trades against live P&L.

Student Survey Coordinator | Women in Economics and Policy, Remote — May 2026 – July 2026
- Faced with low engagement on the Annual Student Survey, designed a targeted participation and incentive strategy that lifted response rates nearly 6x.
- Built a Power BI dashboard (Power Query, DAX) to translate survey results into actionable program recommendations, and co-founded the MSE chapter of WiEP.

Undergraduate Teaching Assistant | Madras School of Economics — Jan 2025 – Apr 2026
- Python & Data Science TA (Dr. Arpita Choudhary), 2026 – created homework sets and led tutorials on NumPy, Matplotlib, Pandas, data cleaning, and Python-based finance/economics optimization problems.
- Mathematical Methods II TA (Dr. Poorna Pushkala Narayan), 2025 – solved weekly problem sheets and supported students' understanding of core quantitative topics.

Summer Intern | Voluntary Health Services, Chennai — May – Jun 2025
- Conducted a literature review on cost-effectiveness of HIV treatments and supported statistical analysis at the intersection of public health and economics.

Research Intern | Energy Consortium, IIT Madras — May – Jul 2024
- Analyzed agricultural and emissions data under Prof. Santhosh Sahu to assess energy efficiency in Indian fertiliser industries.

RESEARCH EXPERIENCE

A Critique of RL Approaches to Collusion in Oligopoly — May – Dec 2025
- Co-authored a literature review with faculty on deep reinforcement learning and tacit collusion in simulated oligopoly markets (SSRN link).

Efficient Market Hypothesis in Indian Equity Markets — Dec 2024 – Feb 2025
- Designed an event study on NIFTY100 dividend announcements testing semi-strong form market efficiency; presented findings in MSE's Research Seminar Series.

ADDITIONAL INFORMATION

Skills: STATA, MATLAB, LaTeX, MS Office, After Effects, R, Claude Code

Leadership & Volunteering: **Vice President**, MSE Cultural Committee (2025–26) — organized and headed ~30 events, managed a budget of over INR 1,50,000, secured sponsors, and drove 6,000+ social media impressions; volunteer at Besant Memorial Animal Dispensary (shelter care) and Nirvriti-Lit Club, Vidya Sagar (reading sessions for children with disabilities)

Languages: English, Tamil, Hindi, German (A1)